

# Electronics 1: Homework #4

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## 3.26

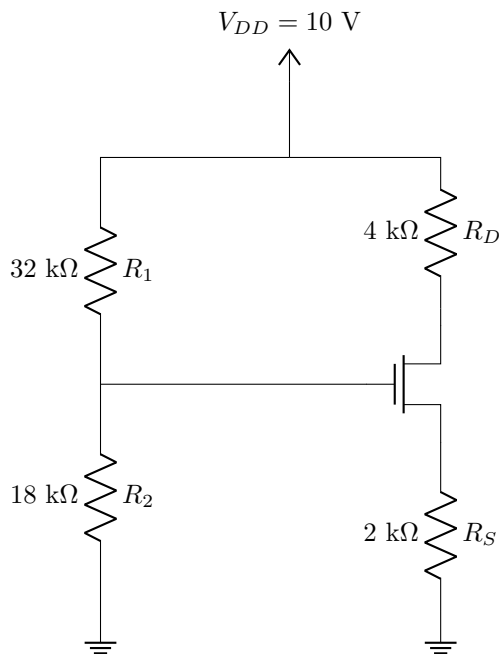


Figure 1: Circuit for Problem 3.26

Given parameters for the NMOS:  $V_{TN} = 0.8$  V,  $K_n = 0.5$  mA/V<sup>2</sup>. Using the voltage divider to find the gate voltage  $V_G$ :

$$V_G = V_{DD} \left( \frac{R_2}{R_1 + R_2} \right) = 10 \left( \frac{18}{32 + 18} \right) = 3.6 \text{ V}$$

The source voltage  $V_S$  is found by:

$$V_S = I_D R_S = I_D (2 \text{ k}\Omega) = 2I_D$$

Thus, the gate-to-source voltage  $V_{GS}$  is:

$$V_{GS} = V_G - V_S = 3.6 - 2I_D$$

Assuming the transistor is in the saturation region:

$$I_D = K_n (V_{GS} - V_{TN})^2$$

$$I_D = 0.5(3.6 - 2I_D - 0.8)^2 = 0.5(2.8 - 2I_D)^2$$

$$I_D = 0.5(7.84 - 11.2I_D + 4I_D^2) = 3.92 - 5.6I_D + 2I_D^2$$

Rearranging into a standard quadratic equation:

$$2I_D^2 - 6.6I_D + 3.92 = 0$$

Solving for  $I_D$ :

$$I_D = \frac{6.6 \pm \sqrt{43.56 - 4(2)(3.92)}}{4} = \frac{6.6 \pm \sqrt{12.2}}{4}$$

The two possible solutions are  $I_D = 2.52 \text{ mA}$  and  $I_D = 0.777 \text{ mA}$ . If  $I_D = 2.52 \text{ mA}$ , then  $V_{GS} = 3.6 - 2(2.52) = -1.44 \text{ V}$ , which implies the transistor is off ( $V_{GS} < V_{TN}$ ). Therefore, this solution is invalid. Using the valid solution  $I_D = 0.777 \text{ mA}$ :

$$V_{GS} = 3.6 - 2(0.777) = 2.046 \text{ V}$$

Now calculate  $V_{DS}$  to verify the saturation region:

$$V_{DS} = V_{DD} - I_D(R_D + R_S) = 10 - 0.777(4 + 2) = 10 - 4.662 = 5.338 \text{ V}$$

Checking saturation condition:

$$V_{DS} \geq V_{GS} - V_{TN} \implies 5.338 \text{ V} \geq 2.046 \text{ V} - 0.8 \text{ V} = 1.246 \text{ V}$$

The assumption holds. The Q-point values are:  $V_{GS} = 2.046 \text{ V}$ ,  $I_D = 0.777 \text{ mA}$ ,  $V_{DS} = 5.338 \text{ V}$ .

## 3.31

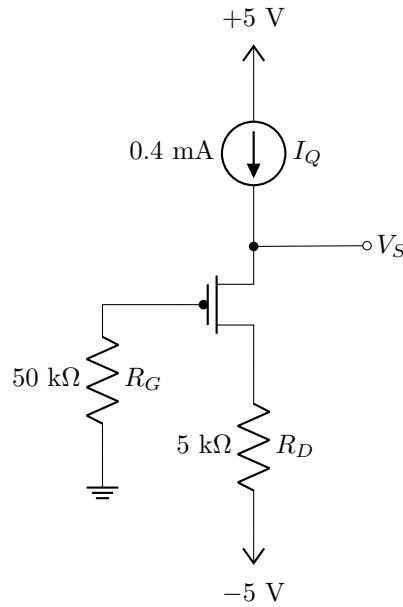


Figure 2: Circuit for Problem 3.31

Given parameters for the PMOS:  $V_{TP} = -0.8$  V,  $K_p = 200 \mu\text{A}/\text{V}^2 = 0.2 \text{ mA}/\text{V}^2$ , and  $I_Q = 0.4$  mA. Since the gate is tied to ground through  $R_G$  and no DC gate current flows ( $I_G = 0$ ), the gate voltage is  $V_G = 0$  V. The drain current  $I_D$  is set directly by the current source  $I_Q$ :

$$I_D = 0.4 \text{ mA}$$

Assume the PMOS is in saturation:

$$I_D = K_p(V_{SG} + V_{TP})^2 \implies 0.4 = 0.2(V_{SG} - 0.8)^2$$

$$(V_{SG} - 0.8)^2 = 2 \implies V_{SG} - 0.8 = \sqrt{2} \approx 1.414 \text{ V}$$

$$V_{SG} = 2.214 \text{ V}$$

Since  $V_G = 0$  V and  $V_{SG} = V_S - V_G$ :

$$V_S = 2.214 \text{ V}$$

To find  $V_{SD}$ , we first determine the drain voltage  $V_D$ :

$$V_D = -5 + I_D R_D = -5 + (0.4 \text{ mA})(5 \text{ k}\Omega) = -5 + 2 = -3 \text{ V}$$

Now we calculate  $V_{SD}$ :

$$V_{SD} = V_S - V_D = 2.214 - (-3) = 5.214 \text{ V}$$

Checking the saturation condition ( $V_{SD} \geq V_{SG} + V_{TP}$ ):

$$5.214 \text{ V} \geq 2.214 \text{ V} - 0.8 \text{ V} = 1.414 \text{ V}$$

The assumption holds. \*\* $V_S = 2.214$  V and  $V_{SD} = 5.214$  V\*\*.

## 3.35

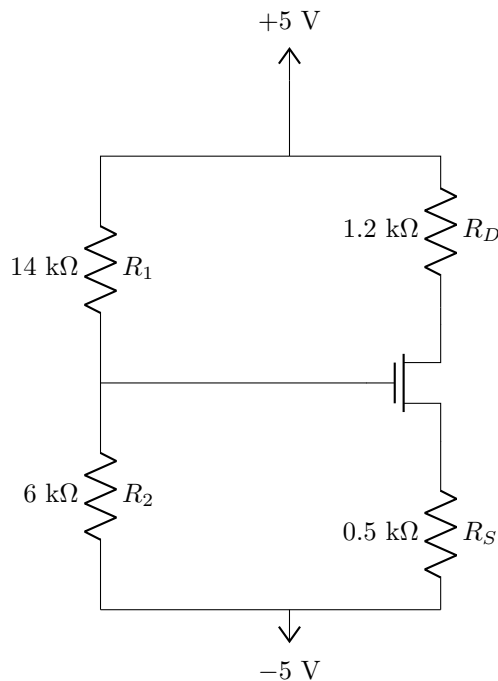


Figure 3: Circuit for Problem 3.35

Given parameters:  $V_{TN} = 0.4$  V,  $k'_n = 120 \mu\text{A}/\text{V}^2 = 0.12 \text{ mA}/\text{V}^2$ ,  $W/L = 25$ . First, calculate  $K_n$ :

$$K_n = \frac{1}{2}k'_n \left( \frac{W}{L} \right) = \frac{1}{2}(0.12)(25) = 1.5 \text{ mA}/\text{V}^2$$

Calculate the gate voltage  $V_G$  via the voltage divider from the 10 V total supply (+5 V to -5 V):

$$V_G = -5 + \left( \frac{6}{14 + 6} \right) \cdot (10) = -5 + 3 = -2 \text{ V}$$

The source voltage is given by:

$$V_S = -5 + I_D R_S = -5 + 0.5 I_D$$

$$V_{GS} = V_G - V_S = -2 - (-5 + 0.5 I_D) = 3 - 0.5 I_D$$

Assuming saturation:

$$I_D = K_n (V_{GS} - V_{TN})^2 = 1.5(3 - 0.5 I_D - 0.4)^2 = 1.5(2.6 - 0.5 I_D)^2$$

$$I_D = 1.5(6.76 - 2.6 I_D + 0.25 I_D^2) = 10.14 - 3.9 I_D + 0.375 I_D^2$$

$$0.375 I_D^2 - 4.9 I_D + 10.14 = 0$$

Solving for  $I_D$ :

$$I_D = \frac{4.9 \pm \sqrt{24.01 - 4(0.375)(10.14)}}{0.75} = \frac{4.9 \pm \sqrt{8.8}}{0.75}$$

$I_D = 10.49$  mA or  $I_D = 2.578$  mA. If  $I_D = 10.49$  mA, then  $V_S = -5 + 0.5(10.49) = 0.245$  V, and  $V_{GS} = -2 - 0.245 = -2.245$  V (Invalid, off). Using  $I_D = 2.578$  mA:

$$V_S = -5 + 0.5(2.578) = -3.711 \text{ V} \implies V_{GS} = -2 - (-3.711) = 1.711 \text{ V}$$

Now calculate  $V_{DS}$ :

$$V_{DS} = V_D - V_S = (5 - I_D R_D) - (-3.711) = (5 - 2.578 \cdot 1.2) + 3.711 = 1.906 + 3.711 = 5.617 \text{ V}$$

Saturation verified ( $5.617 > 1.711 - 0.4$ ). \*\* $V_{GS} = 1.711 \text{ V}$ ,  $I_D = 2.578 \text{ mA}$ ,  $V_{DS} = 5.617 \text{ V}$ \*\*.

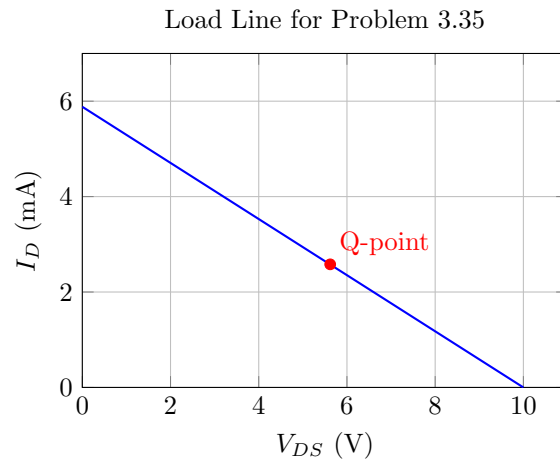


Figure 4: DC Load Line and Q-Point

## 3.43

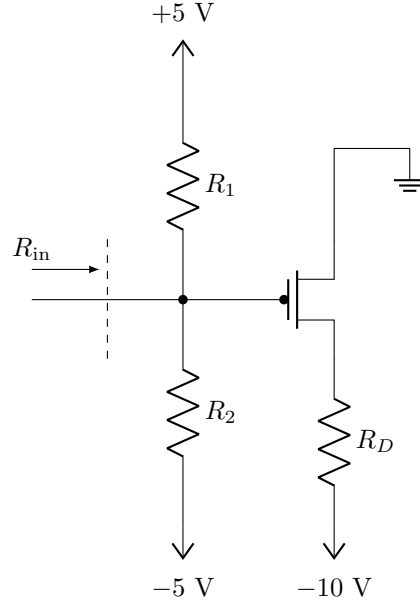


Figure 5: Circuit for Problem 3.43

Given PMOS parameters:  $V_{TP} = -1.75$  V,  $K_p = 3$  mA/V<sup>2</sup>. Design requirements:  $I_D = 5$  mA,  $V_{SD} = 6$  V, and  $R_{in} = 80$  k $\Omega$ .

The source is tied to ground, so  $V_S = 0$  V. For  $V_{SD} = 6$  V:

$$V_{SD} = V_S - V_D \implies 6 = 0 - V_D \implies V_D = -6$$
 V

The drain current flows through  $R_D$  to the  $-10$  V supply:

$$V_D = -10 + I_D R_D \implies -6 = -10 + (5 \text{ mA}) R_D \implies R_D = \frac{4}{5} = 0.8 \text{ k}\Omega = 800 \Omega$$

Assume PMOS is in saturation to find the required  $V_{SG}$ :

$$I_D = K_p (V_{SG} + V_{TP})^2 \implies 5 = 3(V_{SG} - 1.75)^2$$

$$(V_{SG} - 1.75)^2 = 1.667 \implies V_{SG} - 1.75 \approx 1.291 \text{ V} \implies V_{SG} = 3.041 \text{ V}$$

Since  $V_S = 0$  V, the gate voltage must be  $V_G = -3.041$  V.

The gate voltage is established by the  $R_1$  and  $R_2$  voltage divider:

$$V_G = -5 + \left( \frac{R_2}{R_1 + R_2} \right) \cdot (10)$$

Let  $x = \frac{R_2}{R_1 + R_2}$ . Then:

$$-3.041 = -5 + 10x \implies 10x = 1.959 \implies x = 0.1959$$

The input resistance is the parallel combination of  $R_1$  and  $R_2$ :

$$R_{in} = R_1 \parallel R_2 = 80 \text{ k}\Omega$$

Using the relationship  $R_{in} = x \cdot R_1$ :

$$R_1 = \frac{R_{in}}{x} = \frac{80 \text{ k}\Omega}{0.1959} \approx 408.37 \text{ k}\Omega$$

Using the parallel combination to find  $R_2$ :

$$80 = \frac{408.37 \cdot R_2}{408.37 + R_2} \implies R_2 = \frac{80 \cdot 408.37}{408.37 - 80} = \frac{32669.6}{328.37} \approx 99.49 \text{ k}\Omega$$

\*\*Design values:\*\*  $R_D = 0.8 \text{ k}\Omega$ ,  $R_1 = 408.37 \text{ k}\Omega$ ,  $R_2 = 99.49 \text{ k}\Omega$ .